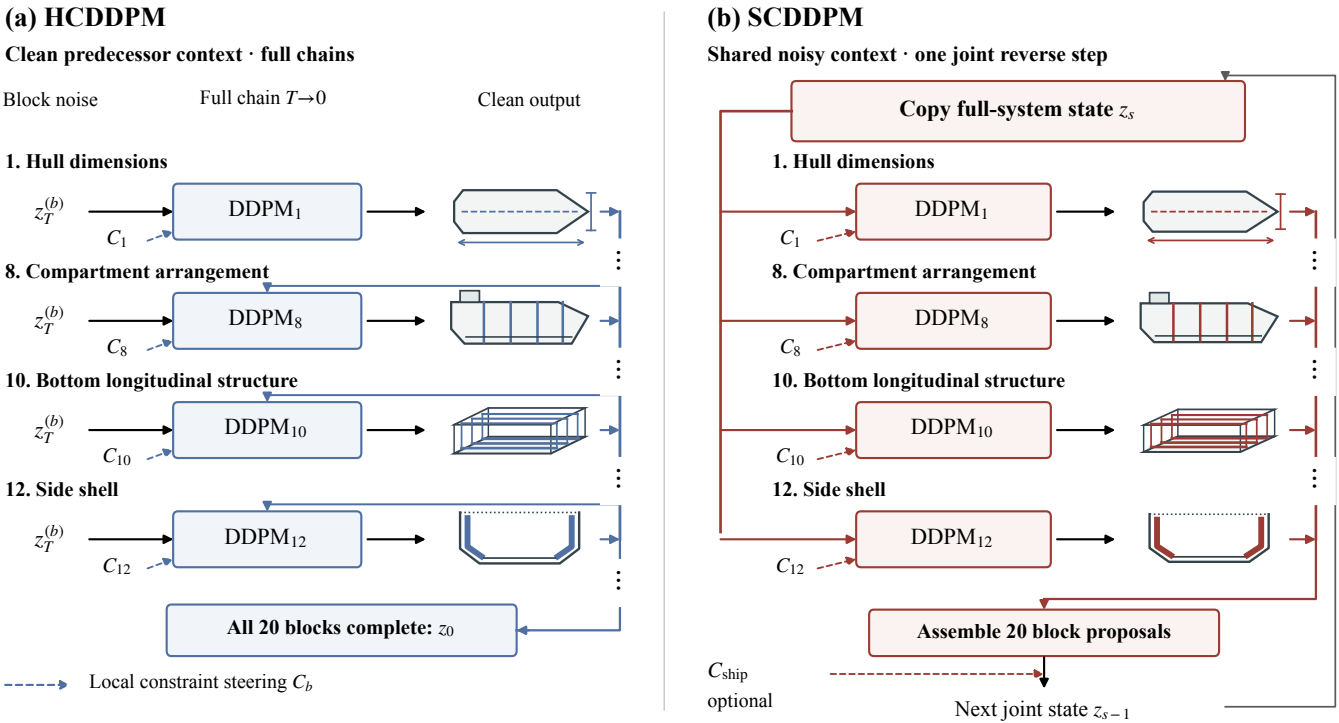

(a) HCDDPM
Clean predecessor context · full chains
Block noise
Full chain $T \to 0$
Clean output
1. Hull dimensions
$z_T^{(b)}$
C_1
DDPM$_1$
8. Compartment arrangement
$z_T^{(b)}$
C_8
DDPM$_8$
10. Bottom longitudinal structure
$z_T^{(b)}$
C_{10}
DDPM$_{10}$
12. Side shell
$z_T^{(b)}$
C_{12}
DDPM$_{12}$
All 20 blocks complete: z_0
Local constraint steering C_b
(b) SCDDPM
Shared noisy context · one joint reverse step
Copy full-system state z_s
1. Hull dimensions
C_1
DDPM$_1$
8. Compartment arrangement
C_8
DDPM$_8$
10. Bottom longitudinal structure
C_{10}
DDPM$_{10}$
12. Side shell
C_{12}
DDPM$_{12}$
Assemble 20 block proposals
C_{ship}
optional
Next joint state z_{s-1}